# DHIMAN GOSWAMI

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## Research Focus

Usable Privacy and Security | Human Computer Interaction | Natural Language Processing | Large Language Models and Generative AI |

Ph.D. candidate working at the intersection of natural language processing, human-computer interaction, and usable privacy. My research answers the question how can moderation systems detect offensive content in transliterated, non-standard writing without exposing the identities of non-offending users. I combine curated multilingual datasets, user studies and interviews, and large language models to build user-centered, privacy-preserving detection systems that protect people from both online harm and unwanted inference.

## 1 Education

**George Mason University**, USA — Jan 2021 - Jun 2027 (Tentative)
Ph.D. in Computer Science

**George Mason University**, USA — Jan 2021 - May 2024
M.Sc. in Computer Science — CGPA: 3.93/4.00

**Bangladesh University of Engineering and Technology (BUET)** — May 2012 - Feb 2017
B.Sc. in Computer Science and Engineering — CGPA: 3.45/4.00

## 2 Work Experience

**Current Position:**

**Graduate Research Assistant**, Data Agency and Security (DAS) Lab, GMU — Dec 2024 - Present
**Graduate Teaching Assistant**, Formal Methods and Models, GMU — Aug 2025 - Present

**Prior Experience:**

**Graduate Research Assistant**, GMU — Jun 2022 - Aug 2025
*Digital Engagement and Empowerment Research (DEER) Lab (Jun-Aug 2025); Language Technology Lab (Jun-Aug 2023); Shehu Lab (Jun-Aug 2022)*

**Graduate Teaching Assistant**, GMU — Jan 2021 - May 2025
*Deep Learning; Machine Learning; Database; Introduction to Computer Programming; Object-Oriented Programming; Human-Robot Interaction*

**Student Wage Employment**, Language Technology Lab, GMU — Jun - Aug 2024

**Lecturer**, Dept. of Computer Science and Engineering, Daffodil International University, Bangladesh — Apr 2017 - Dec 2020
*Courses: Intro to Bioinformatics, Data Structures, Algorithms, OOP, Microprocessor & Assembly Language, Computer Architecture and Organizations, Artificial Intelligence, Social and Professional Ethics of Computing*

## 3 Publications

**Citations: 221** | Conference: 21 published, 3 on arXiv, 2 under review | Journal: 1 published | Poster: 2 Published

### Papers Under Review

- **CHI 2027**: The ACM Conference on Human Factors in Computing Systems
- **IUI 2027**: The Annual ACM Conference on Intelligent User Interfaces

### Published Papers / Posters

1. D. Goswami, Z. Lu, S. Das. *Understanding the Perception of Transliterated Offensive Social Media Posts Among Banglish and Hinglish Users.* SCP 2026: Conference on Sociotechnical Cybersecurity and Privacy.
2. S. Das, D. Goswami, M. Melo, A. Johri, VG Motti. *Teaching Usable Privacy in HCI Education: Designing, Implementing, and Evaluating an Active Learning Graduate Course.* EduCHI 2026.
3. D. Goswami, ANB Emran, MHU Sadi, S. Das. *SoK: Analysis of Privacy Risks and Mitigation in Online Propaganda Detection through the PROMPT Framework.* ASIACCS 2026.
4. D. Goswami, JK Naveen Kumar, S. Das. *NLP Privacy Risk Identification in Social Media (NLP-PRISM): A Survey.* EACL 2026.
5. P. Acharya, JE Leibl, D. Goswami, K. North, M. Zampieri, A. Anastasopoulos. *Tracing L1 Interference in English Learner Writing Using a Longitudinal Corpus with Error Annotations.* EMNLP 2025.
6. ANB Emran, D. Goswami, MHU Sadi, S. Das. *TRIDENT - A Three-Tier Privacy-Preserving Propaganda Detection Model in Mobile Networks using Transformers, Adversarial Learning, and Differential Privacy.* WiSec 2025.

7. D. Goswami, M. Zampieri, K. North, S. Malmasi, A. Anastasopoulos. *Multilingual Native Language Identification with Large Language Models.* NAACL SRW 2025.
8. N. Raihan, D. Goswami, SSC Puspo, ML Siddiq, C. Newman, T. Ranasinghe, JCS Santos, M. Zampieri. *On the Performance of Large Language Models on Introductory Programming Assignments.* Journal of Intelligence Information Systems.
9. D. Goswami, S. Thilagan, K. North, S. Malmasi, M. Zampieri. *Native Language Identification in Texts: A Survey.* NAACL 2024.
10. D. Goswami, K. North, M. Zampieri. *GMU at MLSP 2024: Multilingual Lexical Simplification with Transformer Models.* BEA 2024, co-located with NAACL 2024.
11. D. Goswami, SSC Puspo, MN Raihan, ANB Emran, A. Ganguly, M. Zampieri. *MasonTigers at SemEval-2024 Task 1: An Ensemble Approach for Semantic Textual Relatedness.* SemEval 2024.
12. MN Raihan, D. Goswami, ANB Emran, SSC Puspo, A. Ganguly, M. Zampieri. *MasonTigers at SemEval-2024 Task 9: Solving Puzzles with an Ensemble of Chain-of-Thought Prompts.* SemEval 2024.
13. SSC Puspo, MN Raihan, D. Goswami, ANB Emran, A. Ganguly, Ö. Uzuner. *MasonTigers at SemEval-2024 Task 8: Performance Analysis of Transformer-based Models on Machine-Generated Text Detection.* SemEval 2024.
14. MN Raihan, D. Goswami, A. Mahmud, A. Anastasopoulos, M. Zampieri. *EmoMix-3L: A Code-Mixed Dataset for Bangla-English-Hindi Emotion Detection.* WILDRE 2024, LREC-COLING 2024.
15. MN Raihan, D. Goswami, SSC Puspo, C. Newman, T. Ranasinghe, M. Zampieri. *CSEPrompts: A Benchmark of Introductory Computer Science Prompts.* ISMIS 2024.
16. A. Ganguly, ANB Emran, SSC Puspo, MN Raihan, D. Goswami, M. Zampieri. *MasonPerplexity at Multimodal Hate Speech Event Detection 2024.* CASE Workshop, EACL 2024.
17. ANB Emran, A. Ganguly, SSC Puspo, D. Goswami, MN Raihan. *MasonPerplexity at ClimateActivism 2024.* CASE Workshop, EACL 2024.
18. D. Goswami, SSC Puspo, MN Raihan, ANB Emran. *MasonTigers@LT-EDI-2024: An Ensemble Approach towards Detecting Homophobia and Transphobia in Social Media Comments.* LT-EDI 2024, EACL 2024.
19. D. Goswami, MN Raihan, A. Mahmud, A. Anastasopoulos, M. Zampieri. *OffMix-3L: A Novel Code-Mixed Test Dataset in Bangla-English-Hindi for Offensive Language Identification.* SocialNLP 2023.
20. MN Raihan, D. Goswami, A. Mahmud, A. Anastasopoulos, M. Zampieri. *SentMix-3L: A Novel Code-Mixed Test Dataset in Bangla-English-Hindi for Sentiment Analysis.* SEALP Workshop 2023.
21. MN Raihan, D. Goswami, SSC Puspo, M. Zampieri. *nlpBDpatriots at BLP-2023 Task 1: Two-Step Classification for Violence Inciting Text Detection in Bangla.* BLP 2023, EMNLP 2023.
22. D. Goswami, MN Raihan, SSC Puspo, M. Zampieri. *nlpBDpatriots at BLP-2023 Task 2: A Transfer Learning Approach towards Bangla Sentiment Analysis.* BLP 2023, EMNLP 2023.
23. D. Goswami, N. Sultana, WR Bristi. *Algorithms for String Comparison in DNA Sequences.* IJCCI 2018.

### Accepted Posters / Doctoral Consortium

- S. Khadka, D. Goswami, S. Das. *Who Acts When the User is Gone? Digital Remains, Survivor Claims, and Post-Mortem Governance*, CSCW 2026
- D. Goswami, S. Das. *CLOAK: Cross-Lingual Obfuscation, Anonymization, and K-Noise for Privacy-Preserving Social Media Discourse Analysis.* DFRWS USA 2026.
- S. Khadka, D. Goswami, S. Das. *Privacy-Preserving Compliance Checks on Ethereum via Selective Disclosure.* NDSS 2026.
- D. Das, K. Fatima, D. Goswami, S. Das. *Frequency-Domain Chaotic Watermarking for Image Authentication Against Deepfakes.* NDSS 2026.
- D. Goswami. *Designing Cross-Lingual, User-Centered Privacy and Security for Code-Mixed and Multimodal Social Media Environment.* ICISS 2025.
- D. Goswami, S. Das. *NLP-PRISM: A Framework for Multi-Dimensional Privacy Risk Detection in Social Media Conversations among Migrant and Refugee Communities.* Converge AI: Government Solutions Forum, GMU.
- ANB Emran, D. Goswami, MHU Sadi, S. Das. *Design and Evaluation of a Privacy Preserving Propaganda Detection Tool - TRIDENT.* PETs 2025.
- D. Goswami. *Privacy-Preserving Conversational Agents for Multilingual, Code-Mixed, and Multimodal Social Interactions.* CUI 2025.
- P. Acharya, D. Goswami, K. North, A. Anastasopoulos, M. Zampieri. *Using LLMs to Analyze L1 Interference in a Longitudinal Learner Corpus.* MASC-SLL 2025.

## 4 Presentation, Peer Review, and Program Committee Service

### Selected Talks and Presentations

New Security Paradigms Workshop (NSPW 2026) | Digital Forensic Research Conference (DFRWS USA 2026) | SoK: PROMPT Framework ( Capital-Area Colloquium, GMU) | Privacy Discourse and Emotional Dynamics in Mental Health Information Interaction on Reddit (CHIIR 2026) | Privacy-Preserving Compliance Checks on Ethereum (NDSS 2026) | Frequency-Domain Chaotic Watermarking (NDSS 2026) | Privacy Risks in Multilingual Online Discourse (Capital-Area Colloquium, UMD) | NLP-PRISM (Converge AI: Government Solutions Forum, GMU) | TRIDENT (PETs 2025) | Privacy-Preserving Conversational Agents (CUI 2025)

### Paper Review

EACL 2026 | NordiCHI 2026 | DravidianLangTech 2026 (ACL) | LT-EDI 2026 (ACL) | EACL SRW 2026 | ACL 2025, 2026 | NAACL SRW 2024, 2025 | NAACL 2025 | COLING 2025 | ACL Rolling Review (Aug 2024) | EMNLP 2024, 2025 (one review marked “Great Review,” 2024) | SemEval 2024 | LREC-COLING 2024 | LT-EDI 2024 (EACL) | SIGCSE TS 2024 | CSBW 2022, 2023 (IEEE-BIBM) | PLoS One (2019)

### Associate Chair, Program Committee, Mentor, Volunteer

- Student Volunteer, CSCW 2026
- Program Committee member, ICWSM 2026
- Primary Associate Chair, CHI 2026 Posters; Secondary Associate Chair, CHI 2026
- Shared Task Mentor, BLP 2025 (IJCNLP-AACL)

- Scientific Committee member, LREC 2026
- Primary and Secondary Associate Chair, CSCW 2025
- Program Committee member, COLING 2025

## 5 Awards and Scholarships

- Honorable Mention Recognition for Poster, The $29^{th}$ ACM Conference on Computer-Supported Cooperative Work (CSCW 2026)
- Student Travel Award, Best Poster Award - 26th Annual Digital Forensic Research Conference (DFRWS USA 2026)
- Stipend Award - Privacy Enhancing Technologies Symposium (PETS 2025)
- Distinguished Academic Achievement Award, M.S. in Computer Science, George Mason University (Spring 2024)
- Token of Appreciation from Vice Chancellor for Outstanding Teaching Performance, Daffodil International University (Fall 2019)
- Full Free Merit Scholarship, Higher Secondary Certificate Examination, Notre Dame College (2011)
- Perfect Attendance Certification, Notre Dame College, Bangladesh (2009-2011)
- Full Free Merit Scholarship, Secondary School Certificate Examination, Ideal School and College (2009)
- Talent Pool Scholarship, Junior School Scholarship Examination, Motijheel Area, Dhaka Board (2008)
- Tuition Waiver for excellent result, class 5-10, Ideal School and College (2005-2009)
- Appreciation Award, 1st position (6 times consecutively), class 5-10, Ideal School and College (2005-2009)

## 6 Professional and Co-Curricular Activities

- Graduate Assembly Program Representative (Computer Science), GAPSA, GMU (Jul 2025 - Present)
- President, Bangladeshi Graduate Students Association, GMU (Apr 2025 - Apr 2026)
- Vice-President, Fairfax Campus, GAPSA, GMU (Jul 2024 - Jun 2025)
- Treasurer, Bangladeshi Graduate Students Association, GMU (Apr 2022 - Apr 2023)
- Team Leader, Course Moderation Committee, Dept. of CSE, Daffodil International University (Oct 2017 - Dec 2020)
- Student Advisor, Dept. of CSE, Daffodil International University (Oct 2017 - Dec 2020)
- Member, Organizing Committee, ACM ICPC Asia Dhaka Regional Contest, Bangladesh (2018)
- Thesis / Project Supervisor, Undergraduate Division, Dept. of CSE, Daffodil International University (Oct 2017 - Dec 2020)
- Class Representative, Dept. of CSE (Batch 11, Section A), BUET (May 2012 - Feb 2017)